

Boyu Yao

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RESEARCH INTERESTS

Power system planning, electricity markets, and resource adequacy under physical, operational, market, and policy constraints; critical minerals, energy supply chains, and grid infrastructure; emerging technologies and large-scale data center electricity demand.

EDUCATION

Johns Hopkins University

Aug. 2023 – Present

Ph.D. in Civil and Systems Engineering

Advisor: Prof. Yury Dvorkin

Affiliations: Ralph O'Connor Sustainable Energy Institute (ROSEI); Critical Minerals Research Lab; Electric Power Innovation for a Carbon-free Society (EPICS); Center for Systems Science and Engineering (CSSE); Academic Center for Reliability and Resilience of Offshore Wind (ARROW)

Johns Hopkins University

Jan. 2021 – May 2022

Master of Science in Engineering, Geography and Environmental Engineering

Advisor: Prof. Ben Hobbs

Concentration: Environmental Management and Economics

Tianjin University

Sept. 2016 – June 2020

Bachelor of Engineering, Environmental Engineering

3+2 Program with **University of California, Irvine**, Henry Samueli School of Engineering

Exchange Student in Civil and Environmental Engineering

Aug. 2019 – June 2020

PUBLICATIONS & RESEARCH OUTPUTS

Preprints & Manuscripts Under Review

- Saroj Khanal, Geon Roh, **Boyu Yao**, Abraham Silverman, Dennice Gayme, Charalambos Konstantinou, Jip Kim, Yury Dvorkin: *Shift or Curtail? How Much Data-Center Flexibility Is Worth Depends on the Host Power Grid*. Manuscript under review; *arXiv preprint*, Aug. 2026.
- **Boyu Yao**, Yury Dvorkin: *Grid-Supporting Equipment Supply Chains Constrain the Feasible Pace of Power System Expansion*. Manuscript under review; *arXiv preprint*, Apr. 2026.

Peer-Reviewed Articles

- Owen Reed, **Boyu Yao**, Saroj Khanal, Magdalena Klemun, Yury Dvorkin: *Electricity Prices under Rapid Load Growth: The Case of Data Centers in Virginia*. *Oxford Energy Forum*, Mar. 2026.
- Ming Wei, **Boyu Yao**, Michael Davidson: *Resource Adequacy Under Institutional Constraints and the Low-Carbon Energy Transition in China*. *Energy Policy*, Mar. 2026.
- **Boyu Yao**, Andrey Bernstein, Yury Dvorkin: *Generation Expansion Planning with Upstream Supply Chain Constraints on Materials, Manufacturing, and Deployment*. *IEEE Transactions on Energy Markets, Policy, and Regulation*, Jan. 2026.
- **Boyu Yao**, Hannah Jeong, Mahdi Mehrtash, Bentley Allan, Daniel Ockerman, Yury Dvorkin: *Understanding Supply Chain Constraints for the U.S. Clean Energy Transition*. *Nature Portfolio Journal Clean Energy*, Sep. 2025.
- **Boyu Yao**, Yury Dvorkin: *A Decision Framework for Clean Energy Supply Chain Optimization*. In *Proceedings of the 2025 IEEE Power & Energy Society General Meeting (PESGM)*, IEEE PESGM, Austin, Texas, USA, July 2025.

Technical Reports, Policy Briefs & Datasets

- Saroj Khanal, **Boyu Yao**, Abraham Silverman, Yury Dvorkin: *Valuing Data Center Load Shifting and Curtailment in PJM*. Policy brief, Ralph O'Connor Sustainable Energy Institute (ROSEI), Johns Hopkins University, ROSEI Policy Brief, Sep. 2026.
- Saroj Khanal, **Boyu Yao**, Marisel Villafañe-Delgado, Yury Dvorkin: *ICARUS PJM Dataset (ICARUS-PJM)*. Versioned dataset, GitHub, Sep. 2026.
- **Boyu Yao**, Abe Silverman, Yury Dvorkin: *The Next Data Center Bottleneck Is Grid-Supporting Equipment*. Policy brief, Ralph O'Connor Sustainable Energy Institute (ROSEI), Johns Hopkins University, ROSEI Policy Brief, Apr. 2026.
- Owen Reed, Saroj Khanal, **Boyu Yao**, Dyer Combariza, Magdalena Klemun, Yury Dvorkin: *ICARUS Data Center Dataset (ICARUS-DC)*. Versioned dataset, GitHub, May 2026.
- Yury Dvorkin, Benjamin Hobbs, Ziting Huang, Stephanie Wilcox, **Boyu Yao**: *Energy Resilience and Efficiency in Maryland*. Technical report for the Energy Resilience and Efficiency Working Group, State of Maryland, Maryland EREWG, Dec. 2024.
- Ming Wei, **Boyu Yao**, Chi Gao, Max Dupuy, Michael Davidson: *North China and Northeast China Interprovincial Electricity Trade Brief*. Electricity Market Tracker, Oct. 2023.
- Fikri Kucuksayacigil, **Boyu Yao**, Ming Wei, Chi Gao, Max Dupuy, Michael Davidson: *Northeast China Grid Winter Resource Adequacy Report*. Electricity Market Tracker, July 2023.

RESEARCH & PROFESSIONAL APPOINTMENTS

Resources for the Future

Washington, DC & Remote

Research Scholar, Critical Minerals Research Lab

Sep. 2026 – Jul. 2027

- Participate in the 2026–2027 research cohort, provide feedback on fellow scholars' research, and engage in policy discussions and collaborative research on critical minerals.

Researcher, RFF Subaward Project

Sep. 2026 – Jan. 2027

Supervisors: Karen Palmer and Jenya Kahn-Lang

- Continued research initiated during the summer internship through an RFF-funded subaward to Johns Hopkins University, supported by the Blue Horizons initiative, focusing on resource adequacy, forward contracting, and strategic market behavior.

Research Intern

Jun. 2026 – Aug. 2026

Supervisors: Karen Palmer and Jenya Kahn-Lang

- Conducted research on capacity markets, resource adequacy, and electricity market design by developing and evaluating analytical models and decision-support tools.

Johns Hopkins University

Baltimore, MD

Graduate Research Assistant

Aug. 2023 – Present

Advisor: Prof. Yury Dvorkin, Smart Energy reseARCH (SEARCH) Group

- **Clean Energy Supply Chains:** Designed a national framework linking critical material supply chains to U.S. decarbonization targets and clean-energy deployment feasibility.
- **Supply-Chain-Constrained Capacity Expansion:** Built an optimization model to quantify how critical material, manufacturing, and land constraints shape generation capacity expansion.
- **Grid-Supporting Equipment Supply Chains:** Integrated stock-flow, bill-of-materials, supply-use, and expansion modeling to assess how equipment and material constraints shape power system expansion.
- **Maryland Energy Resilience and Efficiency:** As part of the Maryland Energy Resilience and Efficiency Working Group (EREWG), analyzed electricity storage pathways and grid transformation strategies to enhance resilience and support Maryland's 2031 decarbonization targets.
- **Advanced Nuclear Integration to PJM:** In collaboration with Constellation, assessed Gen IV nuclear flexibility and PJM market impacts under rising data center demand using capacity expansion modeling.

Research Assistant

Jan. 2022 – May 2022

Advisor: Prof. Ben Hobbs

- Enhanced the JHSMINE energy optimization model by adding variables and constraints, updating system cost calculations, and automating data processing and visualization.

University of California, San Diego

San Diego, CA

Research Associate

Aug. 2022 – Aug. 2023

Supervisor: Prof. Michael Davidson, Power Transformation Lab

- **Winter Resource Adequacy Analysis:** Conducted power system reliability analyses for the Northeast China Grid, providing insights into resource adequacy, seasonal demand, and renewable energy integration.
- **Interprovincial Electricity Trade Analysis:** Assessed the impact of long-term contracts on system costs and reliability in China's power grid, proposing alternative designs through simulation-based analysis.

Tsintergy Technology

Guangzhou, Guangdong, China

Consultant

May 2021 – Dec. 2021

Energy Operations Research Center, Tsinghua University

- Analyzed Guangdong's energy generation and market mechanisms and developed a carbon emissions calculation guide based on IPCC methodology.
- Supported renewable energy policy analysis and energy demand forecasting for Qinghai and Jiangsu through strategy assessment, MATLAB model enhancement, grey modeling, and emissions estimation.

TEACHING & MENTORING EXPERIENCE

Teaching Experience

- **Teaching Assistant, Energy Systems** — Johns Hopkins University, Fall 2024, Fall 2025, Fall 2026. Instructor: Prof. Yury Dvorkin. Led weekly office hours and graded assignments and exams for 30+ students; supported instruction in unit commitment, economic dispatch, and optimal power flow.
- **Teaching Assistant, Energy Policy and Planning Models** — Johns Hopkins University, Spring 2022. Instructor: Prof. Ben Hobbs. Graded assignments and exams for 20+ students and supported review sessions on energy policy and power system planning models.

Research Mentorship

- **Luca Montoya** — JHU ROSETAS REU, Summer 2025. Resource adequacy assessment for capacity expansion planning.
- **Princeton Nguyen** — JHU ROSETAS REU, Summer 2025. Weather-informed scenario generation for capacity expansion planning in ISO-NE.
- **Maria Vabson** — Johns Hopkins University, Sep. 2023 – May 2024. Critical material identification, stock assessment, and production metrics for clean-energy supply chains.

SELECTED PRESENTATIONS

- **(Poster)** *Grid-Supporting Equipment Supply Chains as a Constraint on Power System Expansion*. 2026 PowerUp Conference.
- **(Invited)** *Beyond Market Signals: Supply-Chain Constraints in Grid Expansion*. 2026 Power Markets Journal Club.
- **(Poster)** *The ICARUS Dataset: An Infrastructure-Centric, Open Dataset for Capacity Expansion, Resource Adequacy, and Data Center Integration*. 2026 Ralph O'Connor Sustainable Energy Institute (ROSEI) Summit.
- **(Poster)** *When the Supply Chain Shapes Long-Term Grid Planning*. 2026 Ralph O'Connor Sustainable Energy Institute (ROSEI) Summit.
- **(Poster)** *When the Supply Chain Shapes Long-Term Grid Planning*. 2025 The Electric Power Innovation for a Carbon-free Society (EPICS) Center US Summit.
- **(Invited)** *Market Signals and Incentives Aren't Sufficient: The Case for Supply-Chain Modeling for Grid Expansion*. 2025 INFORMS Annual Meeting.

- **(Oral)** *When the Supply Chain Shapes the Grid: Integrating Upstream Constraints into Long-Term Planning*. JHU Energy and Environment Systems Seminar, 2025.
- **(Oral)** *Integrating Upstream Supply Chains into Generation Expansion Planning*. JHU Center for Systems Science and Engineering (CSSE) Meeting, 2025.
- **(Poster)** *A Decision Framework for Clean-Energy Supply-Chain Optimization*. 2025 IEEE Power & Energy Society General Meeting.
- **(Poster)** *Understanding Supply Chain Constraints for the U.S. Energy Transition*. 2025 JHU Sustainability Symposium on Research, Practice, and Education.
- **(Poster)** *Understanding Clean-Energy Supply Chains: A U.S. Perspective*. 2025 Ralph O'Connor Sustainable Energy Institute (ROSEI) Summit.
- **(Oral)** *Energy Resilience and Efficiency Analyses to Support the Maryland Climate Change Commission*. Annual Meeting of the Energy Resilience and Efficiency Working Group, Maryland Department of the Environment, 2025.
- **(Invited)** *A Fully Open-Source Power Dataset for Policy Applications in China*. 2023 INFORMS Annual Meeting.
- **(Oral)** *Power System Resource Adequacy in the Low-Carbon Transition—Northeast China Grid Winter Resource Adequacy*. 30th Technical Workshop of the China Energy Modeling Forum (CEMF).

SERVICE

- **Peer Reviewer** — Ongoing. Reviewed manuscripts for *IEEE Transactions on Energy Markets, Policy and Regulation*, *IEEE Access*, *Mitigation and Adaptation Strategies for Global Change*, *BMC Environmental Science*, etc.
- **Graduate Orientation Mentor** — Johns Hopkins University, Civil and Systems Engineering, 2024 – Present. Mentored incoming graduate students during departmental orientation and their transition into the program.
- **Volunteer Science Teacher** — Chiang Mai, Thailand, Oct. 2017. Developed and taught science lessons for primary school students.

AWARDS & HONORS

- **RFF Blue Horizons Research Subaward** (*Resources for the Future; Johns Hopkins University*) **2026–2027**
- **IEEE PES General Meeting Travel Fellowship** (*IEEE Power & Energy Society*) **2025**
- **Joseph Meyerhoff Fellowship** (*Johns Hopkins University*) **2023**
- **The 20th “Students of Science & Technology Outstanding Talents”** (*Tianjin University*) **2019**
- **Youth “Eco-Innovation Green Action” Competition** (*National, Third Prize*) **2019**
- **The 15th “Challenge Cup” National Undergraduate Curricular Academic Science & Technology Competition** (*Provincial, First Prize*) **2019**
- **The 12th “Challenge Cup” National Undergraduate Curricular Academic Science & Technology Competition** (*Tianjin University, First Prize*) **2019**
- **The 4th Tianjin College Students Environmental Discipline Innovation & Practice Invitational Tournament** (*Provincial, Second Prize*) **2018**
- **Representative of 3rd Plenary of the Fifth Council, Chinese Society for Engineering Education (CSEE)** (*“Integration of New Engineering Construction with Industry and Education” Int’l Conference*) **2018**
- **Merit Student** (*Tianjin University*) **2017**